

paper yoyos quasi-static model

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1 Simplified geometric model

The system can be modelled as a constant-width rectangular ribbon of negligible thickness wrapped around a uniform-radius cylinder at a uniform pitch. One end of the ribbon is fixed in place to prevent rigid body motion, while the other (“active” end) is displaced in the z direction (the axis of the cylinder). Refer to Figure 1 for a visual representation of the system. The system has the following geometric constants:

- L : the length of the unrolled rectangle.
- W : the width of the unrolled rectangle. Assume $L \gg W$.
- θ_L : the total wind angle of the helix, which is fixed based on the boundary conditions. $\theta_L = 2\pi n$ where n is the number of winds around the z axis (not necessarily integer).

The state of the system is parametrized by the parameter x , which describes how far the active end has displaced in the z direction relative to the base end.

Positions on the helix can be described with local surface u, v coordinates, which describe the unrolled rectangle. The u direction runs along the length of the unrolled rectangle, and the v direction runs across its width. Alternatively, positions can be described in 3d space using cylindrical r, θ, z coordinates.

To simplify the geometry, we assume that the helix has a constant radius r and a constant pitch angle ϕ (the angle the local $\hat{u}$ makes with the r, θ plane). By unrolling the rectangle, we get:

$$\phi = \sin^{-1} \left(\frac{x}{L} \right) \tag{1}$$

$$r = \frac{L}{\theta_L} \cos \phi = \frac{\sqrt{L^2 - x^2}}{\theta_L} \tag{2}$$

2 Quasi-static Equilibrium

We will model the paper yoyo as a 2d membrane. We will ignore forces due to elasticity or plasticity, only friction, and will assume all deformation is developable bending rather than in plane stretching.

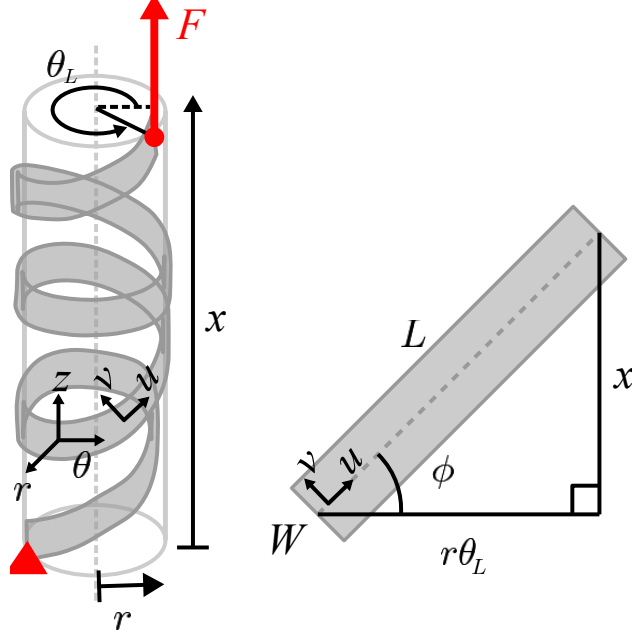


Figure 1: in progress

An area element $du \times dv$ at point (u, v) has a plane stress tensor:

$$\sigma(u, v) = \begin{bmatrix} \sigma_{uu} & \sigma_{uv} \\ \sigma_{uv} & \sigma_{vv} \end{bmatrix} \quad (3)$$

2.1 In-plane force balance

Now we perform a force balance on the area element in u and v . We model the ribbon as a thin physical sheet of uniform thickness $t \ll W \ll L$. An area element of length du and width dv has a volume $du dv t$. Let $\sigma(u, v)$ denote the 3D Cauchy plane-stress tensor acting within the membrane, with components σ_{uu} and σ_{uv} having dimensions of force per unit area (N/m^2). The in-plane membrane forces per unit length (N/m) transmitted across the faces of the differential element are therefore $\sigma_{uu}t$, $\sigma_{uv}t$, and $\sigma_{vv}t$.

The area element is not perfectly flat, but because $\cos \theta \approx 1$ for small angles, we can treat it as flat for the in-plane force balance. Let body forces due to friction f_u, f_v denote the force components per unit volume in the u and v directions. The force balance in u is:

$$[\sigma_{uv}(u, v + dv) - \sigma_{uv}(u, v)] t du + [\sigma_{uu}(u + du, v) - \sigma_{uu}(u, v)] t dv + f_u t du dv = 0 \quad (4)$$

$$\frac{\sigma_{uv}(u, v + dv) - \sigma_{uv}(u, v)}{dv} + \frac{\sigma_{uu}(u + du, v) - \sigma_{uu}(u, v)}{du} + f_u = 0 \quad (5)$$

$$\boxed{\frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{uu}}{\partial u} + f_u = 0} \quad (6)$$

Similarly, the force balance in v is:

$$\boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + f_v = 0} \quad (7)$$

2.2 Out-of-plane force balance

Next, we take the out-of-plane force balance by projecting these membrane forces onto the normal vector of the element at (u, v) , that is, $\hat{r}$. Because the element is wrapped around a cylinder of radius r , the local basis vectors $\hat{u}$ and $\hat{v}$ rotate continuously in 3D space. Evaluating the differential rotation of the Darboux frame over the steps du and dv , the inward radial tilt of the basis vectors produces an outward radial restoring force from the internal Cauchy stresses:

- On the $u + du$ face, $\hat{u}$ rotates inward by an angle $\frac{\cos^2 \phi}{r} du$, contributing a radial force of $-(\sigma_{uu} t dv) \frac{\cos^2 \phi}{r} du$.
- On the $u + du$ face, $\hat{v}$ rotates outward by $-\frac{\sin \phi \cos \phi}{r} du$, contributing $(\sigma_{uv} t dv) \frac{\sin \phi \cos \phi}{r} du$.
- On the $v + dv$ face, $\hat{v}$ rotates inward by an angle $\frac{\sin^2 \phi}{r} dv$, contributing a radial force of $-(\sigma_{vv} t du) \frac{\sin^2 \phi}{r} dv$.
- On the $v + dv$ face, $\hat{u}$ rotates outward by $-\frac{\sin \phi \cos \phi}{r} dv$, contributing $(\sigma_{uv} t du) \frac{\sin \phi \cos \phi}{r} dv$.

We balance the net radial projection of these internal membrane forces against the inward clamping force exerted by overlapping adjacent layers. To express this balance directly in terms of quantities with Cauchy stress dimensions (N/m^2) and eliminate explicit dependence on membrane thickness t , we define the effective normal clamping pressure P . Balancing the forces across the volume element yields:

$$P \left(\frac{t}{r} \right) du dv - \sigma_{uu} t \left(\frac{\cos^2 \phi}{r} \right) du dv - \sigma_{vv} t \left(\frac{\sin^2 \phi}{r} \right) du dv + 2\sigma_{uv} t \left(\frac{\sin \phi \cos \phi}{r} \right) du dv = 0 \quad (8)$$

Dividing through by the common geometric volume factor $\left(\frac{t}{r} \right) du dv$ directly isolates the effective clamping pressure P :

$$\boxed{P = \sigma_{uu} \cos^2 \phi + \sigma_{vv} \sin^2 \phi - 2\sigma_{uv} \sin(2\phi)} \quad (9)$$

2.3 Friction

An area element $du \times dv$ at point (u, v) has a position vector given in cylindrical coordinates by:

$$\vec{p}(u, v, x) = \begin{bmatrix} r \\ \theta \\ z \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{L^2 - x^2}}{\theta_L} \\ \frac{u \cos \phi - v \sin \phi}{r} \\ u \sin \phi + v \cos \phi \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{L^2 - x^2}}{\theta_L} \\ \frac{u \theta_L}{L} + \frac{v x \theta_L}{L \sqrt{L^2 - x^2}} \\ \frac{u x}{L} + \frac{v \sqrt{L^2 - x^2}}{L} \end{bmatrix} \quad (10)$$

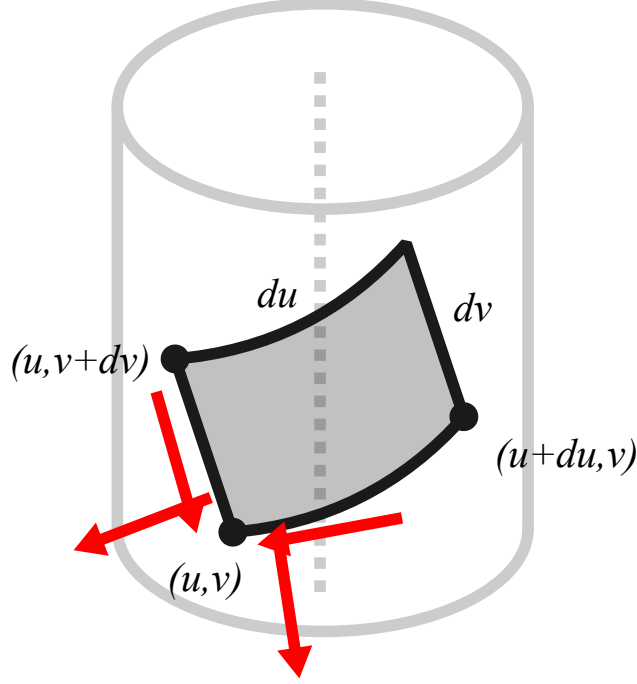


Figure 2: in progress

Then, the velocity of the area element is:

$$\frac{\partial}{\partial x} \vec{p}(u, v, x) = \begin{bmatrix} \frac{\partial r}{\partial x} \\ \frac{\partial \theta}{\partial x} \\ \frac{\partial z}{\partial x} \end{bmatrix} = \begin{bmatrix} -\frac{x}{\theta_L \sqrt{L^2 - x^2}} \\ -\frac{L \theta_L v}{(L^2 - x^2)^{3/2}} \\ \frac{u}{L} - \frac{vx}{L \sqrt{L^2 - x^2}} \end{bmatrix} \quad (11)$$

An area element at point (u, v) overlaps with an interior area element at point (u^-, v^-) if their cylindrical coordinates have equal r and z , but $\theta^- = \theta - 2\pi$. Therefore,

$$u^- = u - 2\pi r \cos \phi \quad (12)$$

$$v^- = v + 2\pi r \sin \phi \quad (13)$$

We make the simplifying assumption that every element has a corresponding interior element, despite the fact that for some elements, v^- may be outside the domain $[-W/2, W/2]$. We can then plug these into (11) to compute the difference in velocity between the two overlapping area elements, which gives the relative slip direction in the r, θ, z frame for an area element at (u, v) :

$$\frac{\partial \vec{p}(u, v, x)}{\partial x} - \frac{\partial \vec{p}(u^-, v^-, x)}{\partial x} = \begin{bmatrix} 0 \\ \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix} \quad (14)$$

As expected, there is no relative velocity in the radial direction (i.e., the layers are not lifting away from each other). We can now compute the relative slip direction in the local u, v frame by

projecting the slip vector onto the local basis vectors. Additionally, we need to convert $\frac{\partial \theta}{\partial x}$ to a linear velocity by multiplying by the radius r :

$$\vec{v}_{slip}(u, v, x) = \begin{bmatrix} r \cos \phi & \sin \phi \\ -r \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix} = \begin{bmatrix} \frac{4\pi x}{L \theta_L} \\ \frac{2\pi(L^2 - 2x^2)}{L \theta_L \sqrt{L^2 - x^2}} \end{bmatrix} \propto \begin{bmatrix} f_u \\ f_v \end{bmatrix} \quad (15)$$

The slip direction is proportional to the friction direction, which gives us a ratio R_f between the friction components in the u and v directions:

$$R_f := \frac{f_u}{f_v} = \frac{2x\sqrt{L^2 - x^2}}{L^2 - 2x^2} = \tan(2\phi) \quad (16)$$

Assume that the Coulomb friction model applies at all points. We can then compute the magnitude of the friction components in u, v based on the out-of-plane normal force:

$$\mu P = \sqrt{f_u^2 + f_v^2} = f_v \sqrt{1 + R_f^2} \implies f_v = \frac{\mu}{\sqrt{1 + R_f^2}} P, \quad f_u = \frac{\mu R_f}{\sqrt{1 + R_f^2}} P \quad (17)$$

$$\boxed{f_u = \frac{\mu P}{r} \sin(2\phi), \quad f_v = \frac{\mu P}{r} \cos(2\phi)} \quad (18)$$

3 Compatibility

The system contains three unknown stress components ($\sigma_{uu}, \sigma_{vv}, \sigma_{uv}$). Because the two in-plane equilibrium equations (Eqs. 6 and 7) are statically indeterminate for three stress fields, we must introduce a third governing equation that enforces the kinematic continuity of the ribbon as a developable 2D membrane.

For a continuous, simply connected sheet undergoing small deformations without tearing or overlap, the in-plane strain components must satisfy the Saint-Venant compatibility condition:

$$\frac{\partial^2 \epsilon_{uu}}{\partial v^2} + \frac{\partial^2 \epsilon_{vv}}{\partial u^2} - 2 \frac{\partial^2 \epsilon_{uv}}{\partial u \partial v} = 0 \quad (19)$$

Assuming linear isotropic elasticity under plane stress, the constitutive relations (Hooke's Law) relate the strain tensor ϵ to the Cauchy stress tensor σ via Young's modulus E and Poisson's ratio ν :

$$\epsilon_{uu} = \frac{1}{E} (\sigma_{uu} - \nu \sigma_{vv}), \quad \epsilon_{vv} = \frac{1}{E} (\sigma_{vv} - \nu \sigma_{uu}), \quad \epsilon_{uv} = \frac{1 + \nu}{E} \sigma_{uv} \quad (20)$$

Substituting Equation (20) into Equation (19) and multiplying through by E , the elastic modulus cancels out entirely:

$$\frac{\partial^2 \sigma_{uu}}{\partial v^2} - \nu \frac{\partial^2 \sigma_{vv}}{\partial v^2} + \frac{\partial^2 \sigma_{vv}}{\partial u^2} - \nu \frac{\partial^2 \sigma_{uu}}{\partial u^2} - 2(1 + \nu) \frac{\partial^2 \sigma_{uv}}{\partial u \partial v} = 0 \quad (21)$$

To eliminate the mixed shear derivative $\frac{\partial^2 \sigma_{uv}}{\partial u \partial v}$, we differentiate the in-plane equilibrium Equations

tion (6) with respect to u and Equation (7) with respect to v , then add the results together:

$$2\frac{\partial^2\sigma_{uv}}{\partial u\partial v} = -\left(\frac{\partial^2\sigma_{uu}}{\partial u^2} + \frac{\partial^2\sigma_{vv}}{\partial v^2} + \frac{\partial f_u}{\partial u} + \frac{\partial f_v}{\partial v}\right) \quad (22)$$

Substituting this identity into Equation (21) and collecting terms yields the Beltrami-Michell compatibility equation for plane stress in the presence of body forces:

$$\nabla^2(\sigma_{uu} + \sigma_{vv}) + (1 + \nu)\left(\frac{\partial f_u}{\partial u} + \frac{\partial f_v}{\partial v}\right) = 0 \quad (23)$$

where $\nabla^2 = \frac{\partial^2}{\partial u^2} + \frac{\partial^2}{\partial v^2}$ is the 2D Laplacian operator.

To express the divergence of the body force field $\left(\frac{\partial f_u}{\partial u} + \frac{\partial f_v}{\partial v}\right)$ strictly in terms of the stress tensor components, we differentiate the Coulomb friction equations from Equation (18). At any frozen macroscopic extension x , the coil radius $r(x)$ and pitch angle $\phi(x)$ are uniform across the membrane and independent of the local surface coordinates (u, v) . Therefore, the spatial derivative operators act exclusively on the normal clamping pressure $P(u, v)$:

$$\frac{\partial f_u}{\partial u} = \frac{\partial}{\partial u} \left(\frac{\mu P}{r} \sin(2\phi) \right) = \frac{\mu}{r} \sin(2\phi) \frac{\partial P}{\partial u} \quad (24)$$

$$\frac{\partial f_v}{\partial v} = \frac{\partial}{\partial v} \left(\frac{\mu P}{r} \cos(2\phi) \right) = \frac{\mu}{r} \cos(2\phi) \frac{\partial P}{\partial v} \quad (25)$$

Adding Equations (24) and (25) together gives:

$$\left(\frac{\partial f_u}{\partial u} + \frac{\partial f_v}{\partial v} \right) = \frac{\mu}{r} \left[\sin(2\phi) \frac{\partial P}{\partial u} + \cos(2\phi) \frac{\partial P}{\partial v} \right] \quad (26)$$

Similarly, we evaluate the spatial derivatives of P based on Equation (9). Because the trigonometric coefficients are again constant with respect to (u, v) , differentiating P with respect to each coordinate simply differentiates the underlying stress fields:

$$\frac{\partial P}{\partial u} = \cos^2 \phi \frac{\partial \sigma_{uu}}{\partial u} + \sin^2 \phi \frac{\partial \sigma_{vv}}{\partial u} - \sin(2\phi) \frac{\partial \sigma_{uv}}{\partial u} \quad (27)$$

$$\frac{\partial P}{\partial v} = \cos^2 \phi \frac{\partial \sigma_{uu}}{\partial v} + \sin^2 \phi \frac{\partial \sigma_{vv}}{\partial v} - \sin(2\phi) \frac{\partial \sigma_{uv}}{\partial v} \quad (28)$$

We now substitute Equations (27) and (28) back into the body force divergence (Eq. 26), and multiply the entire expression by the elastic term $(1 + \nu)$ from Equation (23). This directly yields the closed-form Beltrami-Michell compatibility equation in terms of the three Cauchy stress components:

$$\boxed{\nabla^2(\sigma_{uu} + \sigma_{vv}) + \frac{\mu(1 + \nu)}{r} \sin(2\phi) \left[\cos^2 \phi \frac{\partial \sigma_{uu}}{\partial u} + \sin^2 \phi \frac{\partial \sigma_{vv}}{\partial u} - \sin(2\phi) \frac{\partial \sigma_{uv}}{\partial u} \right] + \frac{\mu(1 + \nu)}{r} \cos(2\phi) \left[\cos^2 \phi \frac{\partial \sigma_{uu}}{\partial v} + \sin^2 \phi \frac{\partial \sigma_{vv}}{\partial v} - \sin(2\phi) \frac{\partial \sigma_{uv}}{\partial v} \right] = 0} \quad (29)$$

4 PDE formulation and solution

We now combine Equations (6), (7), and (29) to get the following coupled system of partial differential equations for the plane-stress components $\sigma_{uu}(u, v)$, $\sigma_{uv}(u, v)$, and $\sigma_{vv}(u, v)$ at any frozen macroscopic extension x :

$$\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \underbrace{A_u \sigma_{uu} + B_u \sigma_{uv} + C_u \sigma_{vv}}_{f_u} = 0 \quad (30)$$

$$\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \underbrace{A_v \sigma_{uu} + B_v \sigma_{uv} + C_v \sigma_{vv}}_{f_v} = 0 \quad (31)$$

$$\nabla^2(\sigma_{uu} + \sigma_{vv}) + A_c \frac{\partial P}{\partial u} + B_c \frac{\partial P}{\partial v} = 0 \quad (32)$$

where P is given by Equation (9). By substituting Equation (9) directly into the Coulomb friction equations (18), the geometric friction coupling coefficients are constant with respect to the local spatial coordinates (u, v) and are given in closed form by:

$$A_u(x) = \frac{\mu}{r} \cos^2 \phi \sin(2\phi), \quad B_u(x) = -\frac{2\mu}{r} \sin^2(2\phi), \quad C_u(x) = \frac{\mu}{r} \sin^2 \phi \sin(2\phi) \quad (33)$$

$$A_v(x) = \frac{\mu}{r} \cos^2 \phi \cos(2\phi), \quad B_v(x) = -\frac{\mu}{r} \sin(4\phi), \quad C_v(x) = \frac{\mu}{r} \sin^2 \phi \cos(2\phi) \quad (34)$$

$$A_c(x) = \frac{\mu(1+\nu)}{r} \sin(2\phi), \quad B_c(x) = \frac{\mu(1+\nu)}{r} \cos(2\phi) \quad (35)$$

Along the lateral boundaries of the unrolled ribbon ($v = \pm W/2$), the long edges are traction free. This imposes the exact longitudinal and transverse stress-free boundary conditions:

$$\sigma_{uv} \left(u, v = \pm \frac{W}{2} \right) = 0 \quad \text{and} \quad \sigma_{vv} \left(u, v = \pm \frac{W}{2} \right) = 0 \quad (36)$$

Rather than assuming local transverse stress gradients vanish arbitrarily, we define the macroscopic width-averaged stress fields $\bar{\sigma}_{ij}(u)$:

$$\bar{\sigma}_{uu}(u) = \frac{1}{W} \int_{-W/2}^{W/2} \sigma_{uu}(u, v) dv, \quad \bar{\sigma}_{uv}(u) = \frac{1}{W} \int_{-W/2}^{W/2} \sigma_{uv}(u, v) dv \quad (37)$$

Integrating the axial equilibrium Equation (30) across the ribbon width W and dividing by W yields:

$$\frac{d\bar{\sigma}_{uu}}{du} + \frac{1}{W} \int_{-W/2}^{W/2} \frac{\partial \sigma_{uv}}{\partial v} dv + A_u \bar{\sigma}_{uu} + B_u \bar{\sigma}_{uv} = 0 \quad (38)$$

By the Fundamental Theorem of Calculus, the integral of the transverse gradient evaluates directly at the lateral boundaries:

$$\int_{-W/2}^{W/2} \frac{\partial \sigma_{uv}}{\partial v} dv = \sigma_{uv} \left(u, +\frac{W}{2} \right) - \sigma_{uv} \left(u, -\frac{W}{2} \right) = 0 - 0 = 0 \quad (39)$$

Thus, the free-edge boundary condition (Eq. 36) analytically eliminates the transverse divergence term from the macroscopic axial balance.

Applying the same width-averaging procedure to Equation (31) and noting that $\sigma_{vv}(u, \pm \frac{W}{2}) = 0$, the Beltrami-Michell compatibility equation becomes decoupled from the effective 1D membrane forces. The governing system collapses into a closed pair of ordinary differential equations for the width-averaged stresses $\bar{\sigma}_{uu}(u)$ and $\bar{\sigma}_{uv}(u)$:

$$\frac{d\bar{\sigma}_{uu}}{du} + A_u \bar{\sigma}_{uu} + B_u \bar{\sigma}_{uv} = 0 \quad (40)$$

$$\frac{d\bar{\sigma}_{uv}}{du} + A_v \bar{\sigma}_{uu} + B_v \bar{\sigma}_{uv} = 0 \quad (41)$$

To decouple the ODEs, we assume $A_v(x) \neq 0$ (which holds for all intermediate extensions where $\cos(2\phi) \neq 0$) and solve algebraically for σ_{uu} using Equation (41):

$$\sigma_{uu} = -\frac{1}{A_v} \left(\frac{d\sigma_{uv}}{du} + B_v \sigma_{uv} \right) \quad (42)$$

Differentiating Equation (42) with respect to u and substituting both expressions back into Equation (40) yields a single second-order, linear, homogeneous ODE for the shear stress field $\sigma_{uv}(u)$:

$$\frac{d^2 \sigma_{uv}}{du^2} + (A_u + B_v) \frac{d\sigma_{uv}}{du} + (A_u B_v - A_v B_u) \sigma_{uv} = 0 \quad (43)$$

The characteristic polynomial of Equation (43) is $r^2 + (A_u + B_v)r + (A_u B_v - A_v B_u) = 0$. By the quadratic formula, the two longitudinal eigenvalues governing the capstan amplification simplify to:

$$\lambda_{1,2}(x) = \frac{1}{2} \left[-(A_u + B_v) \pm \sqrt{(A_u - B_v)^2 + 4A_v B_u} \right] \quad (44)$$

Assuming a strictly positive discriminant, the general closed-form solutions for the Cauchy stress fields along the ribbon are:

$$\sigma_{uv}(u) = K_1 e^{\lambda_1(x)u} + K_2 e^{\lambda_2(x)u} \quad (45)$$

$$\sigma_{uu}(u) = -\frac{1}{A_v} \left[K_1 (\lambda_1(x) + B_v) e^{\lambda_1(x)u} + K_2 (\lambda_2(x) + B_v) e^{\lambda_2(x)u} \right] \quad (46)$$

where K_1 and K_2 are integration constants (N/m²) determined by the base clamping pre-tension at $u = 0$.

Finally, to derive the relationship between the global extension x and the macroscopic pulling force measured by the Instron, we integrate the projected Cauchy stresses across the cross-sectional

area ($W \times t$) at the active pulling edge ($u = L$). Using $\sin \phi = x/L$ and $\cos \phi = \sqrt{L^2 - x^2}/L$:

$$F(x) = \int_{-\frac{W}{2}}^{\frac{W}{2}} t [\sin(\phi) \sigma_{uu}(L) + \cos(\phi) \sigma_{uv}(L)] dv \quad (47)$$

$$= \frac{Wt}{L} \left[x \sigma_{uu}(L) + \sqrt{L^2 - x^2} \sigma_{uv}(L) \right] \quad (48)$$

Substituting Equations (45) and (46) into the boundary integral and grouping the geometric projection factors into macroscopic amplitude prefactors (N):

$$M_i(x) = \frac{WtK_i}{L} \left[\sqrt{L^2 - x^2} - \frac{x}{A_v(x)} (\lambda_i(x) + B_v(x)) \right] \quad \text{for } i \in \{1, 2\} \quad (49)$$

the macroscopic Instron force collapses into the exact closed-form double-exponential capstan equation:

$$\boxed{F(x) = M_1(x)e^{\lambda_1(x)L} + M_2(x)e^{\lambda_2(x)L}} \quad (50)$$

5 Experimental validation

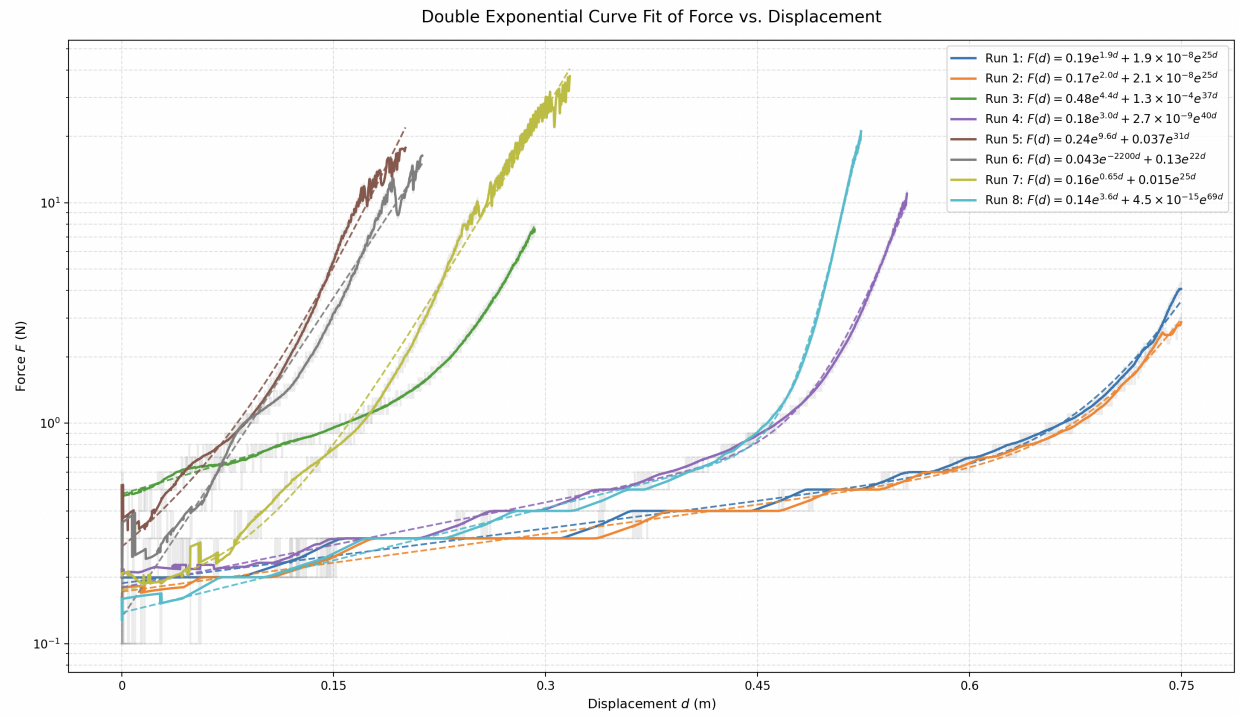


Figure 3: Instron force-displacement data, with raw quantized data (grey), smoothed data (solid colored), and curve fit (dashed colored).